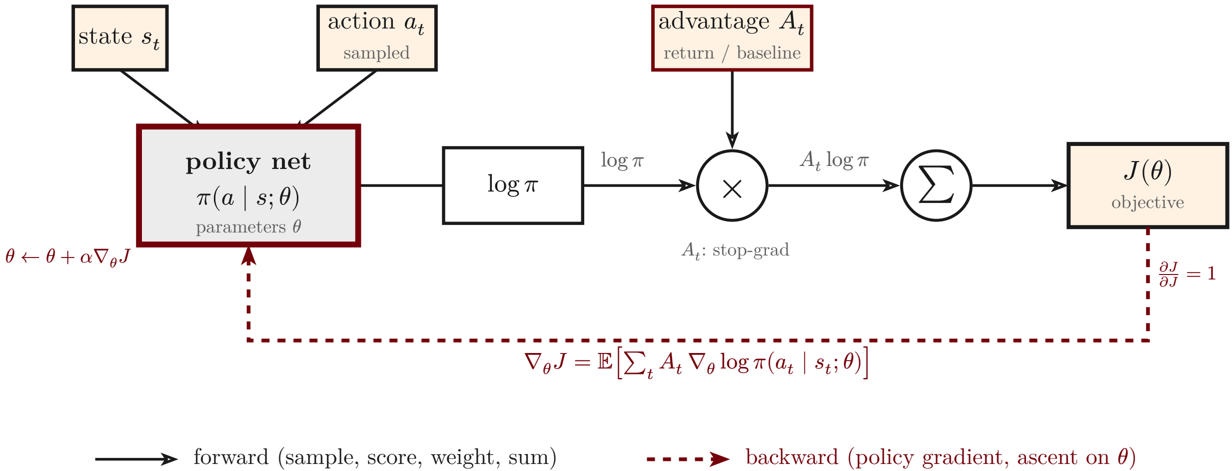
Policy gradient (REINFORCE): the score-function estimator

$$J(\theta) = \mathbb{E}\Big[\textstyle\sum_t A_t \cdot \log \pi(a_t \mid s_t;\theta)\Big], \qquad \nabla_\theta J(\theta) = \mathbb{E}\Big[\textstyle\sum_t A_t \cdot \nabla_\theta \log \pi(a_t \mid s_t;\theta)\Big]$$



The estimator weights each action's log-probability by its advantage A_t and sums over the trajectory. Differentiating the surrogate $J(\theta)$ — with A_t held constant (a stop-gradient) — yields the **score function** $A_t\, \nabla_\theta \log \pi$: ascending it raises the probability of actions with positive advantage and lowers it for negative ones.